

FOX_Checklist

What this is: every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

Last updated: 21 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection (`chore/foxbyte`)

Source: `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

✔ Done & tested — automated tests (unit, VM integration or CI)

🔸 Partial — gaps, or not verified end to end

Sub-features: ✔ built 🔸 partly ❑ not built yet ✖ broken · "VM §n" = section of `make integration` ; "v2 §n" = `make integration-v2` ; "update" = `make integration-update`

✔ Done — tested by hand only

❌ Known issue

❑ Not started

Where we are

Area	Features	Done & tested	Done (manual)	Partial	Known issue	Not started	Progress
A · Install & setup	9	4	3	2	·	·	78%
B · CLI & services	2	2	·	·	·	·	100%
C · Branching & storage	8	6	1	1	·	·	88%
D · Durability	4	3	·	1	·	·	75%
E · Security foundations	3	3	·	·	·	·	100%
F · SQL gateway	7	6	1	·	·	·	100%
G · Accounts & auth	5	4	·	1	·	·	80%
H · Blackbox (record of schema changes)	18	17	·	1	·	·	94%
I · Import & migration	13	5	8	·	·	·	100%
J · ETL pipelines (experimental)	4	2	2	·	·	·	100%
K · APIs, SDKs & agents	6	4	2	·	·	·	100%
L · Web console	11	1	10	·	·	·	100%
M · Build, CI & releases	6	5	1	·	·	·	100%
Total (features built or started)	96	62	28	6	·	·	94%
N · Not started (roadmap)	11	·	·	·	·	11	0%

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:18-bookworm` image plus `wal-g` (`postgres:16-bookworm` for an install created before 21 Sep 2026). This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

Suggested next steps

- Run `fox update` against a real release on macOS and Windows (A9) — cut v0.9.0, install it, then update to v0.9.1.
- HA beyond the demo: archive WAL while main runs on the promoted standby, then automatic failover and fencing (D4). (*Safety fixes C3, D4 hazards, H7, I5, I6 done.*)
- Verify downloads everywhere: installers and `fox setup`'s engine download still skip `SHA256SUMS` (A1, A2, A7).

4. ~~Check the web pages that were built but never used signed in: Integrity, Policies, sidebar (L8–L10); add Blackbox filters (L5).~~ Done — both, in the browser.
5. Add automated tests where there are none: imports (I1–I3, I7 — I5 and I8 now covered), SDKs (K3, K4), web UI (L*).
6. Run the VM integration suites in CI (M2).

A Install & setup

7 of 9 done

ID	Feature · sub-features	Tested by
 A1	One-line installer (macOS / Linux) ✓ Detects OS / CPU, downloads the release binary ✓ Installs to the prefix (sudo only if needed) ✓ Stages the Linux engine for the VM on macOS	Unit (<code>test_install_sh.sh</code> , CI); manual
 A2	Windows installer (install.ps1) ✓ Installs WSL, elevates, resumes after reboot ✓ Downloads fox.exe, engine, image build context ✓ Ubuntu rootfs verified against Ubuntu's checksums ✓ Adds to PATH and runs <code>fox setup</code>	Pester (Windows CI); Windows run 16 Sep
 A3	fox setup on macOS (Lima) ✓ Creates / starts the Lima VM ✓ Installs Docker and ZFS ✓ Reinstalls the engine on every run ✓ Starts the stack	Manual (from-scratch VM run)
 A4	fox setup on Windows (WSL2 + btrfs) ✓ Imports a dedicated distro with systemd ✓ Docker, btrfs and a storage mount unit ✓ Quiet steps, detail in install.log ✓ Waits for systemd on a cold distro	Unit (<code>host_wsl_test.go</code>); Windows run 16 Sep
 A5	Linux native ✓ Engine runs in-process, no VM	VM integration (all suites run on Linux)
 A6	Command forwarding & host file streaming ✓ macOS → Lima VM, starts the VM if stopped ✓ Windows → WSL distro	Unit (<code>TestImportLocalFile</code> , <code>TestWSLArgs</code>)
 A7	Always-latest engine binary ✓ Downloads the newest engine into the cache (ELF check) ✓ <code>FOX_NO_REFRESH</code> , <code>FOX_VERSION</code>	Unit (<code>host_test.go</code> , <code>update_test.go</code>); Windows run 16 Sep
 A8	New-release notice on fox start ✓ Background check, 1.5 s limit, printed last ✓ Only a promoted, newer, complete release	Unit; update \$1, \$1b, \$2, \$3, \$8; real Mac check
 A9	fox update ✓ Downloads and verifies everything before changing anything ✓ Stops servers, installs engine, restarts old one on failure ✓ <code>up</code> , Blackbox upgrade, <code>start</code> on the new engine ✓ Replaces host <code>fox</code> , keeps previous binaries	Unit; update \$4–\$7; a real Mac run

B CLI & services

2 of 2 done

ID	Feature · sub-features	Tested by
----	------------------------	-----------

✓	B1 The fox command set	✓ Stack: setup, start, stop, up, down, status, logs, psql, update ✓ Branching, backup / restore, HA ✓ Blackbox, policy, impact ✓ Import, pipelines ✓ Users, API keys, admin	✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM) □ <code>branch create --from</code> ✓ <code>fox uninstall</code> removes it all, retired installs included	Unit (<code>main_test.go</code> , <code>update_test.go</code>); VM suites
✓	B2 Background services (daemon)	✓ Control plane :8080, gateway :6432, agent API :8088 ✓ Detached, pidfiles and logs in <code>~/ .fox</code>	✓ Stop (TERM then KILL), status, logs □ Health check / restart on crash	Unit (<code>TestStopTerminatesProcess</code>); VM §1

C Branching & storage

7 of 8 done

ID	Feature · sub-features	Tested by
✓	C1 Storage drivers (ZFS / btrfs) ✓ ZFS snapshot + clone per branch (macOS VM, Linux) ✓ btrfs subvolume snapshots (Windows) ✓ Pool / filesystem created on a loopback file	Unit; VM §2, §2b (ZFS); btrfs manual on Windows
✓	C2 Branch lifecycle ✓ Create, list, delete ✓ Reset to parent ✓ Suspend / resume ✓ Recreate a container from its surviving data	Unit; VM §2, §2b, §4
✓	C3 Protection of main ✓ Delete and reset refused	Unit; VM §4, §6
✓	C4 Private networking (no host ports) ✓ Branch databases reachable only through the gateway	Unit (<code>TestParsePublishedPort</code> , <code>TestDSN</code>)
✓	C5 Scale-to-zero & wake on connect ✓ Idle branches suspended automatically ✓ Next connection wakes the branch (~0.3–0.7 s)	VM §7
🔗	C6 Storage quotas ✓ Space reserved for main (ZFS) 🔗 Per-branch cap (opt-in)	Manual
✓	C7 Capacity & status ✓ Services, main health, backups, branches ✓ Storage used / available	VM §1
✓	C8 Provisioning & image pull ✓ Docker, storage pool and image prepared on up	Manual

D Durability

3 of 4 done

ID	Feature · sub-features	Tested by
✓	D1 WAL archiving to object storage ✓ wal-g ships main's WAL to MinIO, off the commit path	VM §3
✓	D2 Base backups ✓ Create and list ✓ Follow the primary after a failover; listing works with main stopped	VM §3, §3b, §6

✓	D3 Point-in-time restore	✓ Restore to a timestamp or latest on :5433 ✓ Per-install credentials ✓ Starts from the newest base backup that precedes the target	✓ A point no backup precedes is refused, naming the oldest	Unit; VM §3, §3b, §6; v2 §1a
🔥	D4 HA standby & failover	✓ Streaming hot standby ✓ Manual failover, <code>main</code> rerouted ✓ Nothing deletes or stops the promoted standby ✓ <code>fox up</code> brings it back, not the old main	✓ <code>fox ha failback</code>, keeping every write ✓ The promoted standby archives WAL, so backups and PITR cover the failover ❑ Automatic failover, fencing, multi-host	Unit; VM §6, §7

E Security foundations 3 of 3 done

ID	Feature · sub-features	Tested by
✓ E1	Per-install generated secrets ✓ Generated Postgres and MinIO passwords ✓ Used by restore too	Unit (indirect); v2 §1a
✓ E2	TLS everywhere ✓ Self-signed TLS on all three services, zero setup 🔥 TLS optional at the gateway (<code>sslmode=disable</code> still connects)	Unit (<code>TestReadStartupUpgradesTLS</code>)
✓ E3	Keys are made by a person, not by start ✓ Start creates no account and no key, and caches none on disk ✓ The banner shows the connection string and how to make a key ✓ The first account created gets the admin grant on main	Integration §1

F SQL gateway 7 of 7 done

ID	Feature · sub-features	Tested by
✓ F1	Routing by database name ✓ One endpoint :6432, database name = branch ❑ Connection pooling, rate limiting	Unit; VM §1, §6
✓ F2	API-key authentication ✓ <code>key_</code> key as the Postgres password ✓ Bad key rejected with a Postgres error	VM §1
✓ F3	Backend login (SCRAM) ✓ Cleartext, md5, SCRAM-SHA-256 ✓ Clients never see the database password ❑ Unit test for SCRAM	VM §1, §6, §8 (exercised)
✓ F4	Per-user Postgres login roles ✓ Role per user, actor can't be forged ✓ Permissions of non-superuser <code>db_client</code>	Manual
✓ F5	Attribution session settings ✓ Actor kind, branch, session id per connection 🔥 Kind / branch / session client-settable	VM §8
✓ F6	Ctrl-C (CancelRequest) forwarding ✓ Cancel a running query through the gateway	Unit
✓ F7	Errors & insecure build tag ✓ Real Postgres error responses ✓ No-auth bypass impossible in release builds	Unit (<code>TestSendError</code>)

G Accounts & auth 4 of 5 done

ID	Feature · sub-features	Tested by
✓ G1	Auth store (SQLite) ✓ Users, keys, sessions, OAuth ids, pipelines ✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once ❑ Roles / permissions, multi-tenant projects	Unit (<code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code>); VM §1

✓	G2 Email & password users ✓ Create account, log in □ Password change / reset	□ List and delete users	Unit; VM §1
✓	G3 API keys ✓ Create (shown once), list, revoke ✓ Bearer / X-API-Key and gateway password	○ Revoking an unknown id reports success	Unit; VM §1
✓	G4 Web sessions & signup control ✓ httpOnly session cookie	✓ Signup can be closed	Unit (TestSessions)
○	G5 OAuth (GitHub / Google) ✓ Sign-in flows built ✓ Defaults match what the engine serves (https://localhost:8080 ; web origin = public URL)	✓ Session cookie Secure / SameSite follow the origin □ Tested with a real OAuth app	Unit; VM §1b

H Blackbox (record of schema changes)

17 of 18 done

ID	Feature · sub-features	Tested by
✓	H1 DDL recording ✓ Who, tool, branch, statement, status, risk ✓ Inherited by every branch	□ Role / database changes, some dropped object types VM §8
○	H2 Hash chain, append-only & verify ✓ Rows can't be edited or deleted normally ✓ fox ledger verify ○ Superuser rewrite caught only via anchors (H10–H12)	✓ bb.policy changes recorded (append-only, not chained) VM §8; v2 §0, §6b
✓	H3 Attribution ✓ Humans via their login role ✓ Agents via their own agent-<id> role	✓ Kind derived from the role, so neither can pose as the other ✓ Tool and session are client-declared by design, and labelled so Unit; VM §8; v2 §1c
✓	H4 Destructive-DDL guardrail ✓ DROP TABLE / DROP SCHEMA refused, attempt recorded ✓ Override only superusers / db_admin , use recorded	✓ "flag" and "allow" actions; unknown actions refused ✓ No hang when blocked after a write in the same transaction VM §8; v2 §1b, §6b
✓	H5 Risk flags ✓ Type change, drop column flagged ✓ Judged on the running statement (no comments or literals)	✓ Automated test Unit; v2 §6b
✓	H6 Blackbox access (CLI / REST / MCP / UI / SDK) ✓ CLI, REST with filters, MCP, web page, SDKs	□ Per-user access control v2 §0, §5
✓	H7 fox ledger revert --to ✓ Help and output say it's a point-in-time restore of main	✓ Points to H13 for a branch from before a change Unit (help text); VM §3 (the restore)
✓	H8 Override permission (db_admin) & upgrade ✓ fox admin grant revoke list ✓ First user granted automatically	✓ fox blackbox upgrade [--all] ✓ Grant / revoke from the web (admins only) v2 §1b, §6
✓	H9 Blackbox 2.0 capture ✓ Transaction id and WAL position per entry ✓ Task, tool-call hash, override use	✓ Fail-safe, append-only ✓ Kill switch not settable by clients Unit; v2 §2
✓	H10 Checkpoints & anchors ✓ Merkle root in the database and in read-only anchor files ✓ On a schedule and on demand	✓ Published anchor format □ Cloud write-once anchors (S3 Object Lock) Unit; v2 §3, §3b
✓	H11 fox-verify (independent verifier) ✓ Checks a live branch or an export	✓ Release binaries for every platform Unit; v2 §3

✓	H12 Blackbox integrity & export	✓ INTACT / TAMPERED via CLI, REST, MCP ✓ JSON-lines export	☐ Streaming for very large ledgers	v2 §3
✓	H13 Branch from just before a Blackbox entry	✓ Restore + replay up to the change's transaction ✓ By transaction id or timestamp	✓ CLI, REST, MCP, web panel ✓ main untouched	Unit; v2 §4
✓	H14 Blackbox names & aliases	✓ <code>fox blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names	✓ Every original name still works	Unit; v2 §5
✓	H15 Policy gate (warn / block)	✓ BBX02 warn / BBX01 block with JSON detail ✓ Blocked attempts recorded ✓ Per-rule <code>db_admin</code> override	✓ CLI, REST, MCP, web page; custom rules ✓ Patterns match the running statement ☐ TRUNCATE	Unit; v2 §6, §6b
✓	H16 Agent provenance & execute_change	✓ Session, task, parent session on every agent change (also through the Gateway DSN) ✓ MCP <code>execute_change</code> with policy preview	✓ <code>fox blackbox sessions</code>, REST 🟡 Provenance is what the agent says (not verified)	Unit; v2 §7
✓	H17 Impact analysis	✓ Dependents (views, foreign keys, indexes, depth) ✓ Other branches that have the object ✓ Policy verdict and risk score	✓ CLI, REST, MCP, inside <code>execute_change</code> 🟡 Functions, types, schemas not recognised from SQL	Unit; v2 §8
✓	H18 Blackbox diff between branches	✓ Fork point, changes only on each side ✓ Objects changed on both	✓ CLI, REST, MCP ☐ Merge (diff only today)	Unit; v2 §8

I Import & migration

13 of 13 done

ID	Feature · sub-features	Tested by
✓ I1	From Postgres ✓ Schema and data via <code>pg_dump</code> ✓ Zero-table result reported as an error	Manual
✓ I2	From MySQL ≤5.7 / MariaDB ✓ <code>pgloader</code> with type mapping (multi-arch image)	Manual
✓ I3	From MySQL 8.x ✓ Native <code>mysqldump</code> path (<code>cached_sha2</code> auth) ✓ Types, NULLs, unicode, PK	Manual
✓ I4	From MongoDB ✓ Collection → typed table, exact name	VM §9, §10
✓ I5	From a .sql dump ✓ Runs the dump into a new branch ✓ Fails listing every failed statement (instance kept)	Unit; VM §11
✓ I6	From JSON / NDJSON ✓ Columns inferred from top-level keys	Unit; VM §10, §11
✓ I7	From CSV ✓ Table from the header, exact column names	Manual

✓	I8 Continuous replication & cutover ✓ Initial copy + streaming (logical replication) ✓ Cutover keeps data (CLI) ✓ Status and cutover from REST and the Import page	✓ A replicating branch is never auto-suspended ✓ Automated test (VM §5b) and driven in the browser	Unit; VM §5b
✓	I9 Guard handling during import ✓ Guardrail paused, recording stays on	✓ Re-armed even after a crash	Manual
✓	I10 Live progress (CLI & SSE) ✓ Live log and progress bar	⚠️ psql subprocess output not streamed	Manual
✓	I11 Import REST endpoints ✓ From a connection string or an uploaded file	☐ Target name validation	Manual
✓	I12 Web Import page ✓ Database and file panels with live console	⚠️ Mentions a SQLite loader that doesn't exist	Manual
✓	I13 Exact identifier fidelity ✓ Case kept (<code>userId</code> , <code>leaseAiChats</code>)	⚠️ Field order of JSON sources not kept	Unit; VM §10

J ETL pipelines (experimental)

4 of 4 done

ID	Feature · sub-features	Tested by
 J1	Pipeline engine ✓ Extract → raw schema → SQL models → tests, in a fresh branch ✓ <code>ref()</code> / <code>source()</code> templating ☐ Dependency graph ☐ Incremental models ☐ Scheduling	Unit; VM §9
 J2	Data-quality tests ✓ <code>not_null</code>, <code>unique</code>, <code>accepted_values</code>, <code>row_count_min</code>, <code>custom</code> ✓ Failure → exit 1, data kept	Unit; VM §9
 J3	Stored pipelines & run history ✓ Saved per user, run over the API, history ☐ Read a run's log ☐ Mark runs whose process died	Manual
 J4	Pipeline editor (web) ✓ Create, edit, run, results, history ☐ Confirm before delete	Manual

K APIs, SDKs & agents

6 of 6 done

ID	Feature · sub-features	Tested by
 K1	Control-plane REST API  Branches, query, Blackbox, policy, impact, import, pipelines, keys  Versioned paths (<code>/api/v1</code>)	Unit; VM §1; v2 §0–§8
 K2	OpenAPI spec  Served, Blackbox 2.0 endpoints documented  Matches the code (import fields, status codes, auth routes, <code>/blackbox</code> aliases)  Drift check in CI	Unit (drift check)
 K3	Python SDK  Branches, query, Blackbox  Import, pipelines, keys, agents  Tests  Publish to PyPI	Manual
 K4	TypeScript SDK  Same client for Node and browsers  Tests  Publish to npm	Manual
 K5	Agent Branch API  A branch per agent, own non-superuser role  Provenance (task, parent, session)  DSN reachable from the host (gateway-routed, branch-scoped key)  Cap, listing and TTL count suspended branches	Unit; VM §5; v2 §1c, §7

✓	K6 MCP server	Unit; v2 §1c–§8
✓	Branches, run_sql, Blackbox tools	✓ Client config & 16-tool reference (docs/mcp.md)
✓	policy_check, execute_change, impact, diff	
✓	Requires an API key and acts as that account; a scoped key is held to its branch	

L Web console 11 of 11 done

ID	Feature · sub-features	Tested by
✓ L1	Embedded single-origin UI ✓ Served by the engine at :8080 ☐ Automated UI tests (whole area)	Manual
✓ L2	Login & signup ✓ Log in / sign up tabs, OAuth buttons	Manual
✓ L3	Dashboard ✓ Live tiles ✓ Branch table: create, suspend, resume, delete, copy DSN	Manual
✓ L4	SQL console & browser ✓ Schema sidebar, rows, structure, indexes ✓ Values shown as Postgres writes them (uuid, interval, inet, bytea) ✓ SQL editor ✓ System tables hidden by default, with a show button ✓ Runs as the signed-in user; one-run override switch ✓ Per-rule override for policy blocks	Manual; API in v2 §6
✓ L5	Blackbox page ✓ Entries per branch with badges ✓ Table / risk / date filters, and "Clear filters" ✓ Verify button; a row expands to its statement and session	v2 §5b; browser
✓ L6	API keys page ✓ Create, copy, list, revoke ✓ A new key is shown inside its connection string (E3's next step)	Manual; browser review 21 Sep
✓ L7	Landing, Guide & Docs pages ✓ Landing, developer guide, reference ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)	Type-check + build
✓ L8	Blackbox integrity page ✓ Verify, checkpoint, export, branch-before ✓ Used signed in; every result matched the CLI ☐ Entry list needs a refresh to show a new change	Type-check + build; browser review 16 Sep
✓ L9	Policies page ✓ Rules, warn/block, custom rules, preview, evaluations ✓ Used signed in; matched the CLI, wrote only to the chosen branch ✓ Non-admin change refused with a readable message ✓ Who can override: list, grant, revoke	Type-check + build; API in v2 §6; browser review 16 Sep
✓ L10	App sidebar layout ✓ Every link opens its page, highlighted, with breadcrumb ✓ Collapse persists across a reload; theme toggle persists ✓ No console errors on any page ☐ Under-860px drawer (needs a manual look)	Type-check + build; browser review 16 Sep
✓ L11	Design system & FoxByte theme ✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values ✓ Ember palette in both themes; filled controls legible (--accent-ink) ✓ Fox mark from theme variables, wordmark from brand.json, favicon ✓ Console, Blackbox, Policies and API keys re-laid out; one table style ✓ Every page reviewed in both themes; console checked at 375px ☐ Not yet reviewed against a running engine with real data	Type-check + build; browser review 21 Sep

M Build, CI & releases 6 of 6 done

ID	Feature · sub-features	Tested by
----	------------------------	-----------

✓	M1 Makefile targets	✓ build, vet, test, integration suites ✓ Suites run only in a throwaway test VM (guarded)	✓ release, web, WSL builds, feature-doc (both PDFs)	CI
✓	M2 Linux CI	✓ Build, vet, test, gofmt ✓ Release script tests, checklist sync check	□ VM integration suites in CI	CI
✓	M3 Windows CI	✓ Build, vet, test, Pester, installer encoding ✓ Guard tests read backslash paths, and generated files are pinned to LF (21 Sep)	□ End-to-end WSL install test	CI
✓	M4 Releases & GHCR image	✓ Binaries for 5 platforms + fox-verify ✓ SHA256SUMS merged safely between jobs	✓ Fails unless fox update 's files are published ✓ Multi-arch Postgres image on GHCR	Script tests in CI
✓	M5 WSL distro & ZFS builds	✓ Prebuilt distro image ✓ Downloaded and verified by the installer (seen on Windows) ✓ Preloaded image tagged as the engine runs it	✓ Imported on a fresh Windows install; preloaded images loaded ✓ Published zstd-compressed, with the gzip size measured beside it in every build log 🟡 zstd unpack on a real Windows install (unit-tested only); the gzip image was 685 MB	Actions artifacts; Pester; Windows run 16 Sep
✓	M6 Postgres image	✓ PostgreSQL 18 + wal-g for a fresh install; one image per supported major (16, 18) ✓ The image follows the data: an install on 16 keeps running 16 after an update ✓ A branch on the wrong major is refused before docker run , with the way out	✓ The major lives in one place, and a test holds the Dockerfile, release matrix and Windows preload to it ✓ All three suites pass on PostgreSQL 18.6: 108 + 277 + 51 checks (21 Sep) □ A command that moves an install from 16 to 18, and a portable backup to take first □ pgvector	Unit; all three suites on 18

N Not started (roadmap)

11 items

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

ID	Item · scope	Related
☐ -	Multi-tenant projects & roles ☐ Projects / organisations ☐ Per-branch permissions (RBAC)	G1, H6
☐ -	Branch merge ☐ Apply one branch's schema changes to another	H18
☐ -	Scheduled backups & retention	D2
☐ -	Automatic HA failover ☐ Health detection, fencing ☐ Multi-host	D4
☐ -	Pipelines: scheduling, incremental / CDC extract, DAG view	J1
☐ -	More import sources: Cassandra, DynamoDB	I1–I8
☐ -	pgvector in the Postgres image	M6
☐ -	Cloud write-once anchors (S3 Object Lock etc.)	H10
☐ -	Release signatures	A9, M4
☐ -	Hosted cloud tier	—
☐ -	Networked storage disaggregation (README roadmap; see the storage engine note on page 1)	C1, D1

Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.